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Threefold Impact on Reaction Coordinate Mapping of Zirconium Based Tri-Chalcogenide Pseudo-Monolayers as Emerging Ultrathin Catalysts for Water Splitting

*Ponnappa Kechanda Prasanna and Sudip Chakraborty**

The continuous quest of finding optimum catalysts for photo-electrocatalytic (PEC) water splitting has been revolving around 2D ultrathin materials because of the fact of exploiting both the surface area. In this work, a threefold impact is envisaged route to enhance the activity both in terms of hydrogen evolution reaction (HER) and oxygen evolution reaction (OER) in the emerging Zirconium Tri-sulfides (ZrS_3) based pseudo-monolayers. Due to the unique and distinct electronic properties of the tri-chalcogenide systems as compared to the di-chalcogenide counterpart of Zirconium, one could afford to explore the implications of not only single-atom functionalization and vacancy defect, but also the external strain to expedite the bifunctional catalytic activity in this promising material. Rigorous electronic structure calculations have been performed for the mentioned threefold effect on not only the band-edge alignment, but going beyond to it by further constructing the reaction coordinate mapping corresponding to HER and OER mechanism. The repercussion of external strain in addition to the single atom functionalization and vacancy defect on the adsorption free energies of the prime intermediates of HER and OER reaction has been further correlated with the work function variation in this pseudo-monolayer system.

1. Introduction

The universal demand for alternative renewable energy resources like solar, hydrogen, wind, hydroelectricity, is prevailing with the need for a clean and green energy demand through the conscious reduction of greenhouse gas emission and global warming. For the sustainable environment, hydrogen with its high energy density is considered as an ideal alternative fuel in the near future with increased energy security.[1,2] Hydrogen (H_2) can be considered as one of the ideal energy carriers with zero carbon emission, which makes it environment friendly and has a high energy density.[3–7] This can be considered as one of the most promising options for replacing fossil fuels, while PEC and electrocatalytic water splitting are considered as the most efficient and sustainable routes for large-scale production of H_2[8,9] as energy carriers. The PEC materials generally use sunlight for the water-splitting mechanism, while in electrocatalysis, external bias has been used. The PEC cell can perform both the HER and the OER, which is the other half of complete water splitting. At the cathode, the reduction reaction occurs, resulting in the production of H_2, while at the anode, the oxidation process occurs, leading to the formation of oxygen O_2. Noble metals are used as electrocatalytic materials for water splitting because of their lower over-potential and rapid reaction kinetics. Currently, materials based on Platinum are the most effective catalysts for HER and Ruthenium and Iridium-based oxides are best known for catalysis for OER.[10–17] These materials are less abundant in nature, which results in substantial production costs. Over the last several decades, a number of investigations have been actively carried out to replace the expensive materials with cost-effective alternates that are abundant in nature and can generate hydrogen.[18–20]

The discovery and exfoliation of graphene have sparked extensive research into synthesizing and predicting new 2D materials.[21,22] Various experimental and theoretical investigations have been carried out on a plethora of 2D materials such as graphene, phosphorene, borophene, silicene, nitrides, transition-metal chalcogenides, etc since then.[23–28] These materials are having larger surface area along with unique electronic, optical, mechanical, transport, and magnetic properties compared to their bulk counterparts.[29,30] The remarkable properties of these materials have been demonstrated in a variety of applications, including photovoltaics, energy storage, sensors, and optoelectronic devices and catalysis.[31–33] Among many promising catalysts, various 2D ultra-thin materials have emerged as promising candidates due to the dual surface exposure of the systems and the increased number of active sites as compared to the bulk counterparts.[34–36] In this regard, the transition-metal-based tri-chalcogenide layered materials are considered to have great potential for catalytic activity from the perspective of HER and OER along with oxygen reduction reaction.[37–39] In 2015, the Group

P. K. Prasanna, S. Chakraborty
Materials Theory for Energy Scavenging (MATES) Lab
Harish-Chandra Research Institute (HRI)
A C.I. of Homi Bhabha National Institute (HBNI)
Chhatnag Road, Jhunsi, Prayagraj, Uttar Pradesh 211019, India
E-mail: sudipchakraborty@hri.res.in

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IV-B transition metal trichalcogenide family of 2D materials was first explored. These materials have a unique chain-like structure in order to have quasi-1D configuration due to their high degree of in-plane lattice symmetry along with its mutually orthogonal crystallographic orientations.[40–43] The general form of these materials is MX_3 (M: Ti, Zr, and Hf; X: S, Se, or Te), while among these materials, newly synthesized ZrS_3 pseudo-monolayer has shown the promise to be considered for vivid applications like photo-luminescence, light-emitting devices, photocatalytic generation of hydrogen peroxide, and thermoelectric.[44–48] In a very recent experimental investigation, ZrS_3 has been explored for multi-wall carbon nanotube supported CuS/ZrS_3 composite for water splitting and dye-sensitized solar cells.[49] It is worth to mention that various methods have been used in the synthesis of these materials, which include chemical vapor transport, solvent thermal process, reversible conversion between ZrS_2 and ZrS_3, and solvent synthesis.[44–46,50–53] Thus, considering the reasonable synthesis process and its application for emerging properties, we have planned to thoroughly investigated the PEC properties of ZrS_3 not only from the band edge alignment perspective, but also from the reaction coordinate mapping of HER and OER mechanism based on systematic electronic structure calculations for the first time to the best of our knowledge. The band edge alignment of this material seems favorable for PEC water splitting. However, in order to be comparable with the highly catalytic noble materials with lower activation barrier,[10] we need to come up with strategies to tune the adsorption free energy corresponding to the adsorption of the primary intermediates of HER and OER mechanism.

Single-atom doping and the formation of defects in the system have been found to improve well-defined catalytic active sites.[54–59] Strain engineering has also been found to play a crucial role in improving catalytic activity.[60] Therefore, in our present work, we have undertaken a three-fold impact strategy to enhance the catalytic activity of ZrS_3 pseudo-monolayer. We have created a S void defect and doping a non-metal element with different atoms such as P, C, and N in the system. Doping these non-metal atoms has found to enhance the catalytic activity in few experiments and computational studies with different monolayers.[57,61–63] Additionally, we have investigated how strain and vacancy defect are having repercussions on the adsorption free energy, which in turn increases the catalytic activity. We have correlated the variation of catalytic activity with the corresponding work function changes. The three-fold impact of single atom functionalization, vacancy defect and external strain on the electronic structure of pristine ZrS_3 pseudo-monolayer and their possible prominent roles corresponding to catalytic activity tuning are investigated scrupulously in this work.

2. Computational Methodology

To understand the water splitting mechanism of Zirconium trisulfides (ZrS_3) pseudo-monolayer system, we have performed a systematic first principles electronic structure calculations within the framework of density functional theory[64,65] (DFT). The Vienna Ab-Initio Simulation Package (VASP) is used to perform all the calculations,[66,67] while the Projector-Augmented Wave (PAW) approach has been used to model the electron-ion interaction in the system. The Perdew–Burke–Ernzerhof (PBE) form of the generalized gradient approximation (GGA)[68] has been used as the exchange correlation functional. The kinetic energy cut-off point for the basis of the plane wave is set at 500 eV. Full ionic relaxation has been performed for all the structures to reach the lowest energy configuration until the Hellman–Feynman forces of the atoms have become less than 1×10^{-3} eVÅ^{-1} and the total energy difference become 1×10^{-4} eV. An adequate thickness of vacuum greater than 7 Å is added on both sides along the z direction to avoid the interaction of the periodic images. The corrections to van der Waals (vdW) interactions are considered within Grimme's DFT-D3 approach.[69,70] The 2D Brillouin zone is sampled using a Monkhorst–Pack[71] k-mesh $5 \times 5 \times 1$ for geometry optimization. We have determined the total and projected density of states (PDOS) and the band structure to study its electronic structure and to study the band alignment of the valence band maximum(VBM) and the conduction band minimum(CBM). In general, the GGA method does not estimate the bandgap accurately,[72] for which, we have considered the Heyd–Scuseria–Ernzerhof type hybrid exchange-correlation functional (HSE06) to determine the bandgap and the corresponding band edge alignment with higher accuracy.[73] To investigate the change in adsorption free energy and the effect of elemental functionalization and defect, a $3 \times 3 \times 1$ supercell is constructed. The same supercell is used to investigate the influence of strain on all the pseudo-monolayer systems. We have determined both the work function and the adsorption free-energy change under the influence of three-fold impact, which further characterize the catalytic efficiency of the system.

3. Results and Discussion

3.1. Crystal Structure Analysis

The primitive cell of ZrS_3 is a rectangular cell, which is having an anisotropic arrangement in both x and y directions and belongs to the P21/M[74] space group. The primitive cell of ZrS_3 comprises 2 Zr atoms and 6 S atoms, with a lattice constant of $a = 5.13$ Å and $b = 3.63$ Å. These values are obtained after full ionic relaxation and are consistent with previously reported results obtained.[44,50,52,75] **Figure 1** represents the optimized geometry of ZrS_3 in a different direction. The monolayer exhibits a chain-like pattern along the y direction and is laterally aligned along the x direction. A close inspection of the structure reveals two distinct types of bonding of the sulfur atom. The sulfur atoms on the surface appear to be in pairs, with a bond length of 2.08 Å, while the internal sulfur atoms are each bonded to one zirconium atom within the chain. The Zr–S bond length within the chain is found to be 2.62 Å which is slightly smaller than Zr–S bond length which is 2.72 Å. An alternately inverted chain of isosceles triangles with equal angles of 73.2°, can be observed along the x axis when viewed in the xz plane.

A void defect is created by removing one of the S atoms and the subsequent atomic functionalization is carried out by replacing one S atom with P, C, or N atom. This results in a doping concentration of 1.39 %. This concentration was determined using Equation (1). The red ball in Figure 1c shows the position

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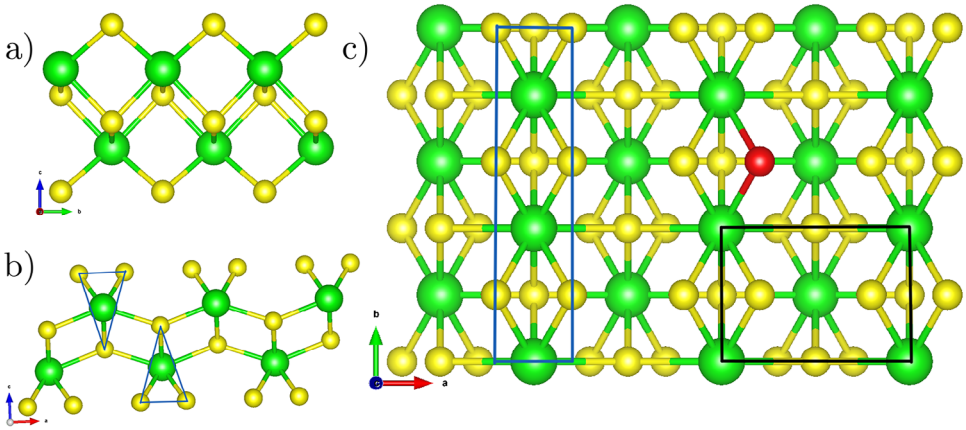

Figure 1. Structure of ZrS_3 along different planes such as a) *yz* plane, b) *xz* plane and c) *xy* plane, the black box represents the unit cell, while the blue box represents the chain-like structure. The green and yellow sphere in the figure represent Zr and S atoms respectively, while the red sphere indicates the position of the void defect or functionalization.

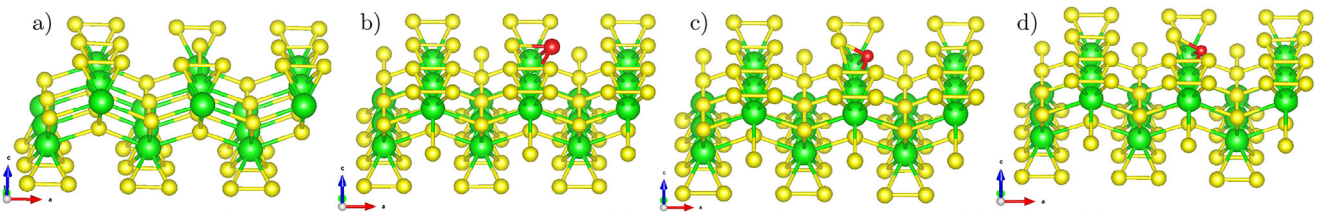

Figure 2. Optimized Geometry of the system after creating a) void defect at S site and functionalization with b) P, c) C and d) N atom.

of the removed atom (void defect) or the replaced atom in the case of functionalization. The optimized geometry of the structure after creating the defect is shown in **Figure 2**. During the doping process, the bond length in and around the doped or defective atom is altered, resulting in anisotropic internal strain. We have also determined the formation energy of all the systems for investigating thermodynamic stability following the equation $E_{form} = (E_{surface} - \sum_i n_i E_i)/\mathrm{N}$, where $E_{surface}$ is the total energy of the pristine or functionalized surface, n_i and E_i are the total number of atom and chemical potential of each atom and N is the total number of atoms. **Table 1** displays the bond length between the constituent atoms and the formation energy of the pristine and functionalized monolayers. The formation energy of all the surface is found to be negative as depicted in Table 1.

Table 1. Lattice parameter of the supercell for a pristine and doped system. X represents the doped atom (S in the case of pristine). Zr–X is the bond length between Zr and the doped atom. The distance between the doped atom and its closest sulfur atom S–X. The distance between the Zr and S atoms at the doped site is Zr–S and Formation energy.

System	$a \times 3$ [Å]	$b \times 3$ [Å]	Zr–X [Å]	S–X [Å]	Zr–S [Å]	E_{form} [eV/atom]
Pristine	15.39	10.89	2.62	2.08	2.62	−1.19
Void defect	15.34	10.89	–	–	2.45	−1.20
P doped	15.38	10.92	2.65	2.08	2.63	−1.17
C doped	15.36	10.87	2.26	1.67	2.82	−1.13
N doped	15.35	10.90	2.16	1.67	2.67	−1.16

$$\text{Doping Concentration} = \frac{\text{Number of dopant atoms or removed S atoms}}{\text{Total Number of atoms present in supercell}} \quad (1)$$

3.2. Electronic Structure Analysis

After finding the optimized configuration for each cases, we have determined the corresponding electronic structure in terms of total and projected density of state of the monolayer ZrS_3 as shown in **Figure 3a**, and the band structure as illustrated in Figure 3b. These are calculated using the HSE06 functional for a better estimation of the bandgap and its alignment. The PDOS of the pristine shown in Figure 3a reveals that the contribution in the valence band (VB) region near the Fermi level is due to the hybridization of the outermost Zr(s) orbital and the outermost S(p) orbital, and the contribution in the conduction band(CB) region near the Fermi region is solely due to the presence of the outermost Zr(s) orbital. The electronic structure of the void defect system and functionalized systems are slightly modified, as shown in Figure S1 of supporting information, which are calculated using GGA exchange-correlation functional. In the case of functionalization we can see a small spike that arises from the p orbital of the doped atom in the VB region near the Fermi level of the system, which may contribute significantly to HER activity. There is no considerable spike that arises from the doped atom in the CB

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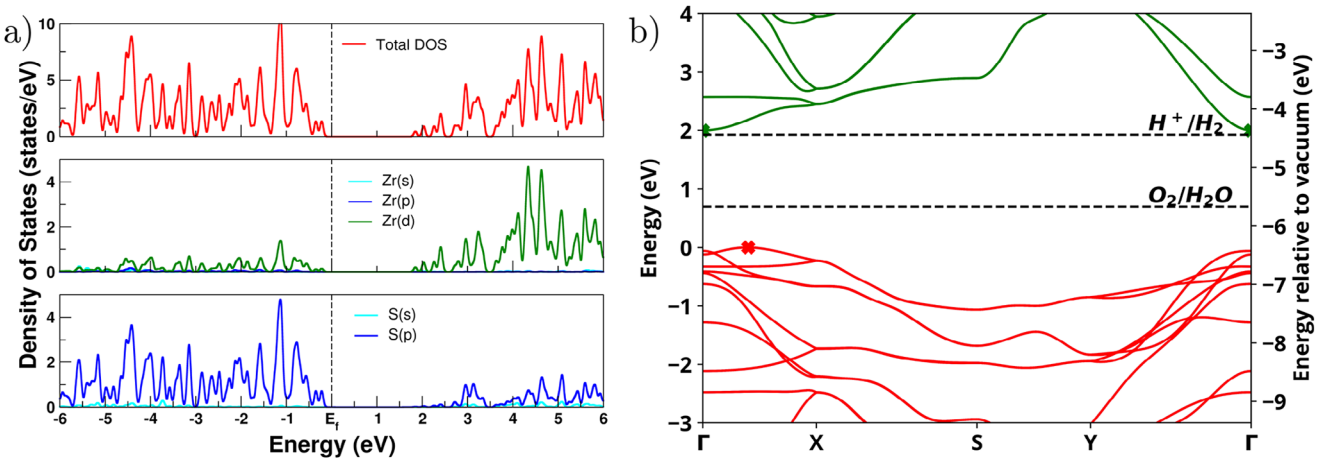

Figure 3. a) Total and projected density of states, and b) band structure along with band alignment with vacuum reference of Pristine surface calculated using the HSE06 functional.

region. However, it influences small changes in the contribution of the S atom.

Understanding the position of CBM and VBM relative to levels H_2O/H_2 and H_2O/O_2 respectively, is essential for the design of the PEC. One of the criteria for the complete PEC water splitting activity of the semiconductor is that the band structure must straddle the redox potential of water.[76,77] This implies that the CBM must be above the H_2O/H_2 level and the VBM must be below the H_2O/O_2 level. On completion of this requirement, we can guarantee that, without any applied external bias, the water-splitting reaction is advantageous when exposed to incident photons. One of the simple approaches is to compute the band edge position of the semiconducting material and the redox levels of water and relate them to any standard reference. The vacuum level is taken as the standard reference point.[78] This allows us to calculate the positions of the edges of the valence and conduction band edges relative to the vacuum potential of the surface. This is done by subtracting the vacuum level from the maximum of the valence band and the minimum of the conduction band, which are obtained from the band structure with the Fermi level shifted to zero. The maximum of the average electrostatic potential along the surface xy is denoted as E_{Vacuum}. The redox potential for water splitting is pH dependent, with the standard reduction potential for H_2O/H_2 being $E_{H_2O/H_2} = -4.44 + pH \times 0.058$ eV and the oxidation potential being $E_{H_2O/O_2} = -5.67 + pH \times 0.058$ eV. At a pH of 0, the standard redox potential and the oxidation potential are −4.44 and −5.67 eV respectively.[79,80]

Figure 3b shows the band structure and band edge alignment. The band structure analysis shows an indirect bandgap of 2.00 eV where the conduction band is located at Γ and the alignment with the vacuum reference is found to be −4.43 eV and the valence band is located between Γ and A where its alignment with the vacuum reference is found at −6.43 eV. The CB corresponds to a potential that is slightly higher than the reduction potential, whereas the VB corresponds to the potential that is significantly lower than the potential needed for OER. These band edges are below and above the oxidation and reduction potentials and therefore, these can be considered as promising candidate for PEC water splitting.

3.3. Reaction Coordinate Mapping of HER and OER Mechanism

The entire process of water splitting involves the HER at the cathode, which is the reduction part, and the OER at the anode, which is the oxidation part. These activities are determined by evaluating Gibbs free energy intermediates using the equation given by Nørskov and co-workers.[30,81,82] The HER mechanism in electrochemical processes are carried out through the reduction of the proton to hydrogen in the cathode. This can be expressed as follows,

$$H^* + H^+ + e^- \rightarrow H_2 \uparrow + * \tag{2}$$

where * represents an active site on the surface and H^* denotes the adsorbed hydrogen atom on the catalytic surface. On the other hand, the complete OER mechanism in the anode is explained by four steps and is called the four-electron pathway. In each step, there is the formation of H^+ and e^-. These elementary reaction steps includes: (i) Water decomposes into hydronium and hydroxide; (ii) hydroxide further breaks down into oxygen and hydrogen; (iii) oxygen then combines with water to form hydroperoxyl; (iv) hydroperoxyl then decomposes to form oxygen. These steps, in the form of equations, can be represented as follows:

$$H_2O + * \rightarrow OH^* + H^+ + e^- \tag{3}$$

$$OH^* \rightarrow O^* + H^+ + e^- \tag{4}$$

$$O^* + H_2O \rightarrow OOH^* + H^+ + e^- \tag{5}$$

$$OOH^* \rightarrow O_2 \uparrow + H^+ + e^- + * \tag{6}$$

In the first step of the OER, the water molecule is adsorbed at the anode material, producing OH^*, which is subsequently adsorbed to give O^*. In the third step, O^* reacts with the water molecule, giving rise to OOH^*, which then releases O_2. Generally, these steps are followed in an acidic medium; even in the presence of an alkaline medium, the primary intermediates remain the same.[83] In our calculation, we have considered the

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acidic medium in which the system is considered to be at pH = 0. The intermediates OH*, O*, and OOH* are involved in different OER steps. The change in adsorption free energy is a major descriptor for determining a better catalytic activity, which is given in Equation (7).

$$\Delta G_{X^*} = \Delta E_{X^*} + \Delta E_{ZPE} - T\Delta S + \Delta G_u \tag{7}$$

where, ΔG_{X^*} is the adsorption free energy and X represents adsorbents, which include H, O, OH, and OOH. The zero-point energy, ΔE_{ZPE}, is estimated to be between 0.1 and 0.4 eV,[81,84] and ΔS is the difference in entropy between the adsorbed state and the gas phase. The values of these parameters are taken from previous studies. $\Delta G_u = eU$, U is the applied electrode potential, e is the elementary charge of an electron. ΔE_X is the adsorption energy which is given by,

$$\Delta E_{X^*} = E_{ZrS_3+X^*} - E_{ZrS_3} - E_X \tag{8}$$

where X^* represents the adsorbate, $E_{ZrS_3+X^*}$ is the total energy of ZrS_3 along with the adsorbate, E_{ZrS_3} is the total surface energy and E_X is the energy of the adsorbate. The adsorption energies of the OER intermediates have been determined following the relationships: $E_{OH} = E_{H_2O} - 1/2E_{H_2}$, $E_O = E_{H_2O} - E_{H_2}$, $E_{OOH} = 2E_{H_2O} - 3/2E_{H_2}$.

The efficiency of the surface to show catalytic activity can be determined by calculating the adsorption free energy changes. For HER activity, the adsorption free energy of adsorbed hydrogen is determined by calculating ΔG_{H^*} as described in Equation (7), while for determining the change in adsorption free energy corresponding to catalytic activity, we have mainly followed Nørskov approach.[81] The ΔE_{ZPE} in the case of HER corresponding to Equation (7) lies in the range 0.01 eV to 0.04 eV. The difference in entropy for hydrogen in the gas phase and the adsorbed state $T\Delta S$ (corresponding to the third term Equation (7) has a value of around 0.4 eV under the experimental conditions.[81] Considering the vibrational entropy of the adsorbed state to be small, implies that the adsorption entropy of $1/2H_2$ is $\Delta S_H \approx -1/2S^0_{H_2}$ where $S^0_{H_2}$ is the entropy of H_2 in the gas phase. Taking this into account, the adsorption free energy is written as $\Delta G_{H^*} = \Delta E_H + 0.24$eV. The optimal value of ΔG_{H^*} should be close to zero for higher activity, which implies that the hydrogen adsorption energy ΔE_H should have a value close to −0.24 eV.

Several possible adsorption sites are considered to find the adsorption site with the minimum energy configuration for hydrogen adsorption. It could be seen that the adsorption of hydrogen on top of S in ZrS_3 corresponds to the minimum energy configuration, as it turns out to be energetically more favorable as compared to the other sites for adsorption. **Figure 4a** shows that the pristine ZrS_3 has a high adsorption free energy change of 1.14 eV, which is less favorable to HER activity. Therefore, we have considered non-metal doping such as P, C, N, and creating void defects in the system to enhance the HER activity through the tuning of adsorption free energy change. During the hydrogen adsorption in the doped and void defect systems, the favorable site for HER activity considering the minimum energy configuration is found to be at the top of the doped site in the case of a doped system near the void defect. The geometries of the system with the minimum energy configurations after hydrogen adsorption are illustrated in the Figure S2 of supporting information. The change in adsorption free energy in the void defect is calculated to be 0.72 eV and the P doped system is −0.69 eV which is found to be enhanced compared to the pristine system. The C and N doped systems are having substantial adsorption free energy change as compared to the pristine system. The system with enhanced HER activity, i.e., the void defect and P doped system along with the pristine system are further subjected to biaxial external strain (both tensile and compressive) with an increment of 2 % up to 6 %. With the application of 8 % compressive strain, we have observed a visible distortion in the geometry of the pristine and P doped system, which could be perceived as structural instabilities. The void defect system undergoes a change in geometry when subjected to a small strain of 4 %. In this study, the change in adsorption free energy has been found to be minimized when compressive strain is applied in the three systems mentioned above, as shown in Figure 4b–4d. It is worth to mention that during the application of compressive strain, the P doped system, which is having the adsorption free energy closer to zero, becomes even better for the corresponding HER activity as compared to the other possibilities. However, there is not much change in adsorption free energy values when tensile strain is applied.

Similar to HER, the adsorption energies for O, OH, and OOH help us determine the OER catalytic activity through the construction of reaction coordinate mapping based on the adsorption free energy values respectively. In the case of O, the corrected adsorption free energy can be approximated as $\Delta G_{O^*} = \Delta E_{O^*} + 0.05$,[85,86] while the correction term for OH and OOH are given by 0.35 and 0.4 eV respectively.[86] Therefore, the adsorption free energy corresponding to OH and OOH can be expressed as $\Delta G_{OH^*} = \Delta E_{OH^*} + 0.35$ and $\Delta G_{OOH^*} = \Delta E_{OOH^*} + 0.4$. The changes in adsorption free energy during the OER mechanism can be defined as $\Delta G_1 = \Delta G_{OH^*}$, $\Delta G_2 = \Delta G_{O^*} - \Delta G_{OH^*}$, $\Delta G_3 = \Delta G_{OOH^*} - \Delta G_{O^*}$, and $\Delta G_4 = 4.92 - \Delta G_{OH^*}$. From the reaction coordinate mapping based on the adsorption free-energy corresponding to OER mechanism, we are able to estimate the magnitude of the overpotential, as mentioned in Equation (9).

$$\eta = \max\{\Delta G_1, \Delta G_2, \Delta G_3, \Delta G_4\}/e - 1.23 \tag{9}$$

The changes in adsorption free energies corresponding to the adsorption of OH*, O*, and OOH* are shown in **Figure 5**. Overpotential of pristine ZrS_3 surface is found to be slightly high, with a value of 1.84 V with the rate-determining step is found to be ΔG_3, which represents the formation of OOH* from O*, whereas the final step for the formation of O_2 from OOH* is observed to be exothermic. Similar to the pristine ZrS_3 the rate determining step for surface with a void defect is G_3, but interestingly overpotential has significantly reduced by 0.68 to 1.16 V. This reduction in overpotential is observed because of the adsorption free energy of O* (ΔG_O) is higher than pristine. It also turns out that ΔG_3 is the rate-determining step with P-doped ZrS_3 for which the overpotential is reduced by 0.2 V compared to the pristine. However in this case the formation of OH* is an exothermic process and both adsorption free energy change for O* and OOH* is reduced resulting in an overall reduction of overpotential. From the comparison of OER activity on the pristine, void and P-doped ZrS_3,

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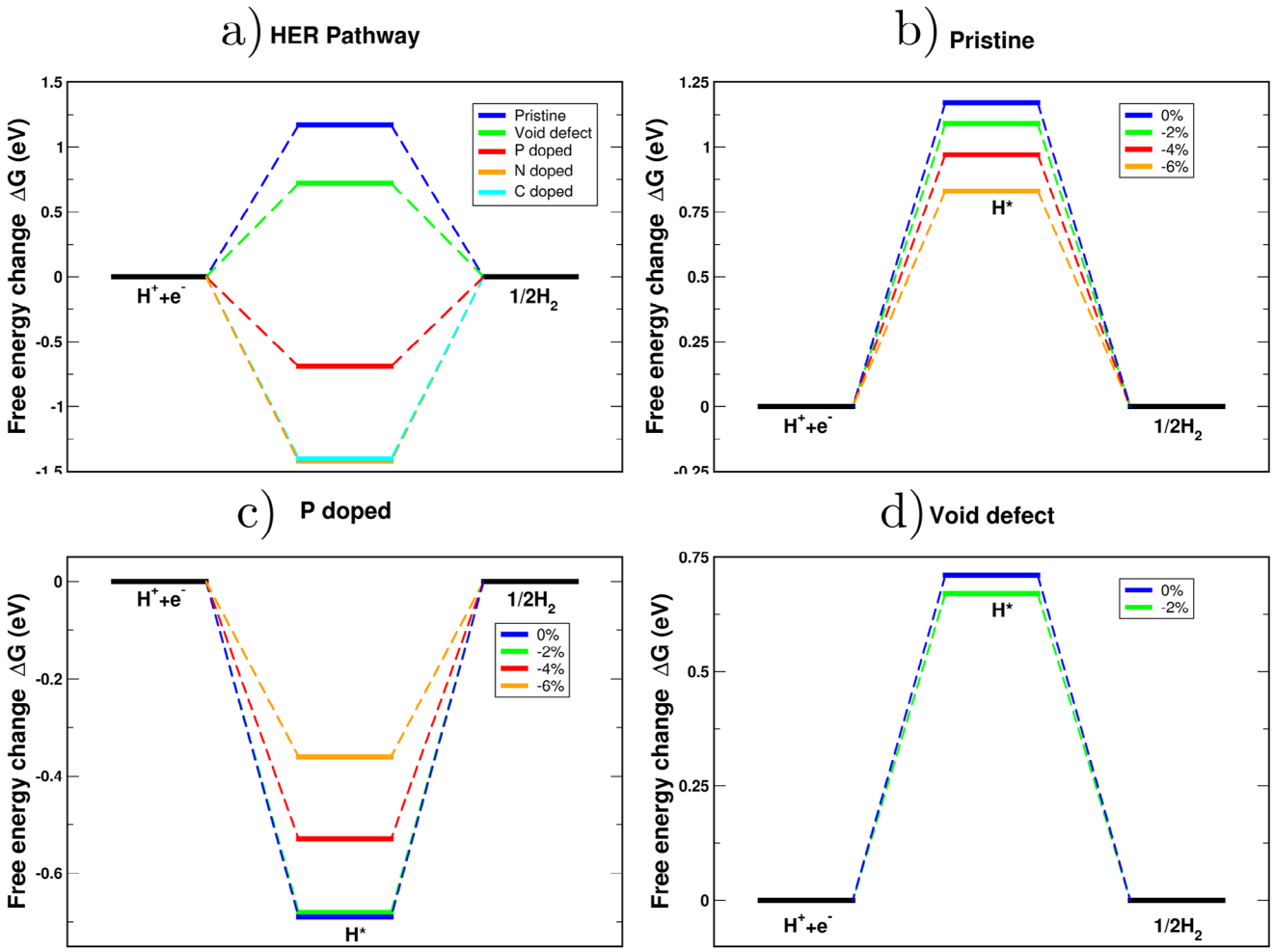

Figure 4. HER activity of ZrS_3 with the effect of doping and compressive strain. a) Pristine, functionalized and void defect system without strain. b) Strain effect on the Pristine surface. c) Strain effect on the P doped surface. d) Strain effect on the surface of the void defect.

we can infer that the void defect with the lower overpotential can emerge as the promising most scenario among them. The introduction of strain to study OER activity is shown in Table S3 (Supporting Information). In pristine ZrS_3 compressive strain reduces the overpotential whereas in the P doped ZrS_3 surface the tensile strain contributes in reducing the overpotential.

3.4. Correlation Between Catalytic Activity and Work Function

The surface properties play always the critical role in catalytic activity. The intermediates such as H^*, O^*, OH^*, and OOH^* formed during the HER and OER mechanism witness the electron transfer between the surface and the adsorbate. The work function

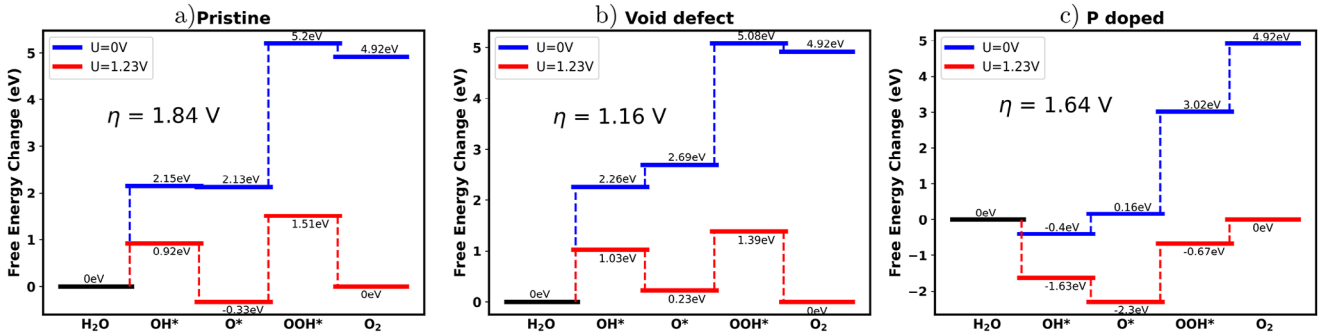

Figure 5. Reaction coordinate mapping corresponding to OER mechanism in a) ZrS_3 with the effect of b) void defect and c) P doping. The red line and blue line corresponds to adsorption free energy profiles under applied overpotential of 0 and 1.23 V respectively.

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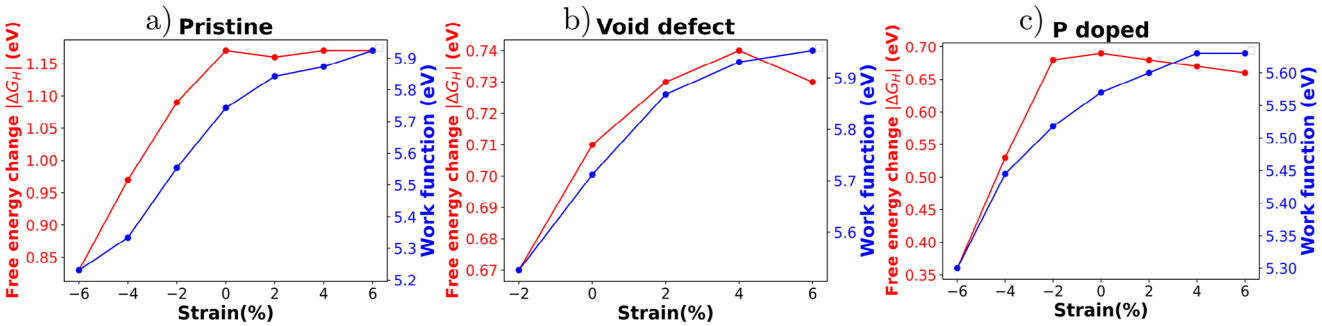

Figure 6. Variation of adsorption free energy in HER and work function as an effect of strain. a) Pristine surface, b) Void defect, c) P doped surface. The blue line corresponds to the values of ΔG_H and the red line corresponds to the work function.

determines the energy required to move an e^- from the surface to be free in a vacuum. In other words, we can say it as the energy required to make e^- free and available for electrocatalytic water splitting. Therefore, the work function of these surfaces is an important parameter to determine the catalytic activity.[87] The work function of the surface can be expressed as the difference between the vacuum and the Fermi level.[88]

$$\Phi = E_{\text{vacuum}} - E_f \tag{10}$$

where, E_{vacuum} is the electrostatic potential of the vacuum level and E_f is the fermi energy of the system. We have studied the variation of the work function with a void defect, doping with P, and pristine, after which the strain has been enforced in these systems. We could observe a good correspondence between the work function and ΔG_H, which is shown in **Figure 6**. The work function has been found to decrease compared to the pristine when creating a void defect and doping with P, showing better efficiency for HER. The effect of strain on the system also follows a similar trend for all three cases, in which the work function gradually decreases when the ΔG_H values move towards zero.

3.5. Correlation Between Catalytic Activity and Charge Transfer Mechanism

We sought to gain a deeper understanding of the interaction between the adsorbed hydrogen atom and the ZrS_3 surface by computing the charge transfer between them. Hence, we have determined the charge density differences between the surface and the adsorbed hydrogen atom, as described in Equation (11).

$$\Delta\rho = \rho_{\text{ZrS}_3+\text{H}} - \rho_{\text{ZrS}_3} - \rho_{\text{H}} \tag{11}$$

where $\rho_{\text{ZrS}_3+\text{H}}$, ρ_{ZrS_3} and ρ_{H} represent the charge densities of ZrS_3 along with the adsorbed hydrogen, ZrS_3 surface and the adsorbed hydrogen. Charge densities for the pristine ZrS_3, void defect, and P-doped surface when the hydrogen atom is adsorbed are depicted in **Figure 7**. The qualitative difference in the charge densities along with the averaged charge density difference along the z-axis is also shown. The yellow area represents the accumulation of charge, whereas the cyan area represents the depletion of charge. In all of the cases, we could see a strong charge-accumulation region near the hydrogen atom that is surrounded by a charge-depletion region around it. For the ZrS_3 surface and void defect system, the peak of the charge accumulation region is in the position of the adsorbed hydrogen atom, which can be visualized from the projection of the differential charge density along the z-axis. The careful observation of the P-doped system showed the peak of charge accumulation at the interface between the surface and the adsorbed hydrogen atom.

Moreover, we also considered the Bader charge of each atom before and after the adsorption of the hydrogen atom. During H adsorption in the pristine system, there is a small change of 0.08e in the S atom near adsorption. However, the S atom has a Bader charge of 0.4e after the H adsorption, which enforces H

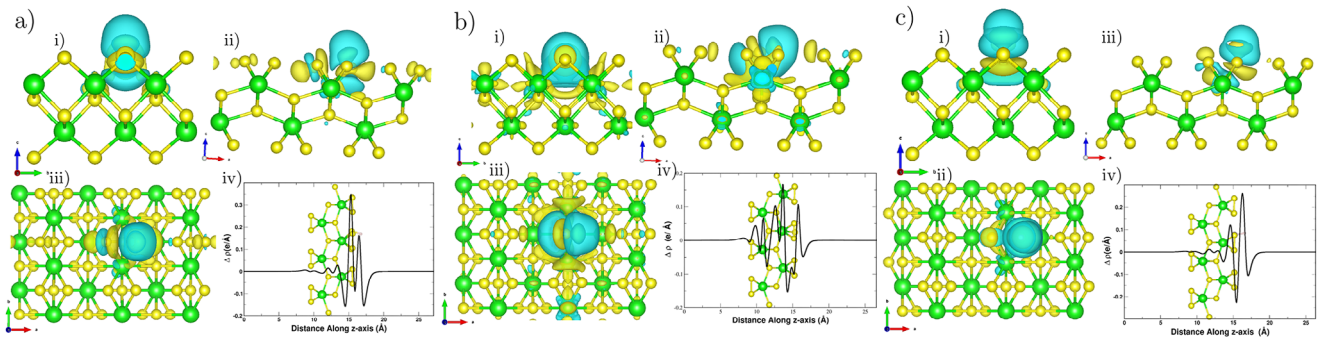

Figure 7. Charge density difference of a) pristine, b) void defect, and c) P doped system. In each of the sub figure the charge density difference along yz plane, xz pane, and xy plane is shown i), ii), and iii) respectively. The projection of charge density difference along the z axis is shown in right bottom corner iv).

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Table 2. Bader charge on the atom near the adsorption site before and after adsorption and charge on the adsorbed atom.

System	Charge on atom near adsorption site [e]		Charge on
	Before adsorption	After adsorption	H atom [e]
Pristine	−0.48	−0.4	0.13
Void	−0.71	−0.62	0.06
P doped	−0.06	0.28	−0.33

to transfer 0.13e. This indicates the loss of electrons around hydrogen, which has also been shown qualitatively. Similarly, to the pristine case, in the void defect the S atom near the void defect and the adsorption site has a change of 0.09e and there is negligible transfer of 0.06e from the H atom. On the other hand, in the P doped monolayer the net charge on the P atom is close to zero(–0.06e) and after the adsorption of H it gains 0.34e, which is a relatively large change compared to the previous cases. This is because 0.33e are transferred from the P atom to the H atom. Due to this stronger charge transfer we obtain a negative adsorption energy in the P doped system, whereas a positive adsorption energy in the pristine and void defect case (**Table 2**).

4. Conclusion

In this work, we have successfully attempted to envisage the threefold impact, namely single atom functionalization, vacancy defect creation and external strain, to synergistically tune the catalytic activity of ZrS_3 pseudo-monolayers. We have not only explored the band edge alignment perspective, but also constructed the reaction coordinate mapping corresponding to HER and OER activities to witness this exciting system as ultrathin catalyst for water splitting. We have correlated the changes in the catalytic activities with the hidden electronic structure analysis in terms of PDOS and band structures, work function variation and the charge transfer mechanism. We have found that void defects and doping with P improve HER activity, while under the influence of external compressive strain, their corresponding hydrogen adsorption free energies move to thermo-neutral condition. We have observed that the work function decreases for systems with a void defect and P functionalization, which are showing better promise for higher HER activity. The effect of strain also shows a similar behavior, with an increase in HER activity and a decrease in work function when compressive strain has been applied. Our Bader charge analysis shows a strong charge accumulation region around the adsorbed hydrogen atom in the promising cases. This work would certainly pave the way to the future research direction, where a synergistic impact of elemental functionalization, vacancy defect and external strain could enhance catalytic activities in pseudo-monolayers, which can be experimentally synthesized.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

catalysis, density functional theory, hydrogen evolution reaction

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[1] S. Bhattacharjee, U. Das, M. Chowdhury, C. Nandi, *Dynam. Energy, Envir. Econ.: A Sustainab. Persp.* **2020**, 115.
[2] J. L. Holechek, H. M. Geli, M. N. Sawalhah, R. Valdez, *Sustainability* **2022**, *14*, 4792.
[3] L. Li, P. Wang, Q. Shao, X. Huang, *Chem. Soc. Rev.* **2020**, *49*, 3072.
[4] M. Cabán-Acevedo, M. L. Stone, J. Schmidt, J. G. Thomas, Q. Ding, H.-C. Chang, M.-L. Tsai, J.-H. He, S. Jin, *Nat. Mater.* **2015**, *14*, 1245.
[5] J. Zhu, L. Hu, P. Zhao, L. Y. S. Lee, K.-Y. Wong, *Chem. Rev.* **2019**, *120*, 851.
[6] K. L. Lim, H. Kazemian, Z. Yaakob, W. W. Daud, *Chem. Eng. Technol.: Indust. Chem.-Plant Equipm.-Proc. Eng.-Biotechn.* **2010**, *33*, 213.
[7] Y. Zheng, Y. Jiao, M. Jaroniec, S. Z. Qiao, *Angew. Chem., Int. Ed.* **2015**, *54*, 52.
[8] K. Zeng, D. Zhang, *Prog. Energy Combust. Sci.* **2010**, *36*, 307.
[9] P. Arunachalam, A. M. Al Mayouf, in Noble Metal-Metal Oxide Hybrid Nanoparticles, 585–606. Elsevier, **2019**.
[10] J. Greeley, T. F. Jaramillo, J. Bonde, I. Chorkendorff, J. K. Nørskov, *Nat. Mater.* **2006**, *5*, 909.
[11] P. Wang, K. Jiang, G. Wang, J. Yao, X. Huang, *Angew. Chem.* **2016**, *128*, 13051.
[12] B. Lim, M. Jiang, P. H. Camargo, E. C. Cho, J. Tao, X. Lu, Y. Zhu, Y. Xia, *Science* **2009**, *324*, 1302.
[13] L. C. Seitz, C. F. Dickens, K. Nishio, Y. Hikita, J. Montoya, A. Doyle, C. Kirk, A. Vojvodic, H. Y. Hwang, J. K. Norskov, T. F. Jaramillo, *Science* **2016**, *353*, 1011.
[14] S. Khamgaonkar, M. Okasha, V. Maheshwari, *Inorg. Chem. Front.* **2023**.
[15] S. Mondal, S. Sarkar, D. Bagchi, T. Das, R. Das, A. K. Singh, P. K. Prasanna, C. Vinod, S. Chakraborty, S. C. Peter, *Adv. Mater.* **2022**, *34*, 2202294.
[16] R. Sadhukhan, A. Kumar, P. K. Prasanna, A. Guha, R. Arenal, S. Chakraborty, T. N. Narayanan, *ACS Appl. Mater. Interfaces* **2024**, *16*, 8627.
[17] I. C. Man, H.-Y. Su, F. Calle-Vallejo, H. A. Hansen, J. I. Martínez, N. G. Inoglu, J. Kitchin, T. F. Jaramillo, J. K. Nørskov, J. Rossmeisl, *ChemCatChem* **2011**, *3*, 1159.
[18] J. Wan, J. Wu, X. Gao, T. Li, Z. Hu, H. Yu, L. Huang, *Adv. Funct. Mater.* **2017**, *27*, 1703933.

 © 2024 Wiley-VCH GmbH

[19] Z. Zhang, G. Yu, H. Li, J. Liu, X. Huang, W. Chen, *Phys. Chem. Chem. Phys.* **2018**, *20*, 10407.
[20] C. Zhang, G. Yu, R. Ku, X. Huang, W. Chen, *Appl. Surf. Sci.* **2019**, *481*, 272.
[21] A. K. Geim, K. S. Novoselov, *Nat. Mater.* **2007**, *6*, 183.
[22] R. Mas-Balleste, C. Gomez-Navarro, J. Gomez-Herrero, F. Zamora, *Nanoscale* **2011**, *3*, 20.
[23] S. Manzeli, D. Ovchinnikov, D. Pasquier, O. V. Yazyev, A. Kis, *Nat. Rev. Mater.* **2017**, *2*, 1.
[24] H. Liu, A. T. Neal, Z. Zhu, Z. Luo, X. Xu, D. Tománek, P. D. Ye, *ACS nano* **2014**, *8*, 4033.
[25] S. Tomar, P. Rastogi, B. S. Bhadoria, S. Bhowmick, Y. S. Chauhan, A. Agarwal, *Europ. Phys. J. B* **2018**, *91*, 1.
[26] E. Bianco, S. Butler, S. Jiang, O. D. Restrepo, W. Windl, J. E. Goldberger, *ACS nano* **2013**, *7*, 4414.
[27] S. Tomar, B. Ghosh, S. Mardanya, P. Rastogi, B. Bhadoria, Y. S. Chauhan, A. Agarwal, S. Bhowmick, *J. Magn. Magn. Mater.* **2019**, *489*, 165384.
[28] C. Yang, X. Li, Y. Liang, *Inorg. Chem. Front.* **2023**.
[29] X. Jiang, A. V. Kuklin, A. Baev, Y. Ge, H. Ågren, H. Zhang, P. N. Prasad, *Phys. Rep.* **2020**, *848*, 1.
[30] Q. Ma, G. Ren, K. Xu, J. Z. Ou, *Adv. Opt. Mater.* **2021**, *9*, 2001313.
[31] K. Drozdowska, A. Rehman, S. Rumyantsev, M. Wurch, L. Bartels, A. Balandin, J. Smulko, G. Cywiński, *Mater. Today Commun.* **2023**, *34*, 105379.
[32] D. Kumar, S. Chakraborty, *Int. J. Hydrogen Energy* **2024**, *51*, 323.
[33] A. Pal, S. Zhang, T. Chavan, K. Agashiwala, C.-H. Yeh, W. Cao, K. Banerjee, *Adv. Mater.* **2023**, *35*, 2109894.
[34] S. H. Mir, S. Chakraborty, P. C. Jha, J. Wärnå, H. Soni, P. K. Jha, R. Ahuja, *Appl. Phys. Lett.* **2016**, *109*, 053903.
[35] Y. Sun, Z. Wang, *Electrocatalysis* **2020**, *11*, 604.
[36] S. Chandrasekaran, L. Yao, L. Deng, C. Bowen, Y. Zhang, S. Chen, Z. Lin, F. Peng, P. Zhang, *Chem. Soc. Rev.* **2019**, *48*, 4178.
[37] Y. Sun, T. Zhang, C. Li, K. Xu, Y. Li, *J. Mater. Chem. A* **2020**, *8*, 13415.
[38] K. Wang, X. Wang, Z. Li, B. Yang, M. Ling, X. Gao, J. Lu, Q. Shi, L. Lei, G. Wu, Y. Hou, *Nano Energy* **2020**, *77*, 105162.
[39] S. A. Younis, E. E. Kwon, M. Qasim, K.-H. Kim, T. Kim, D. Kukkar, X. Dou, I. Ali, *Prog. Energy Combust. Sci.* **2020**, *81*, 100870.
[40] R. Ahammed, A. Rawat, N. Jena, M. K. Mohanta, A. De Sarkar, *Appl. Surf. Sci.* **2020**, *499*, 143894.
[41] J. O. Island, R. Biele, M. Barawi, J. M. Clamagirand, J. R. Ares, C. Sánchez, H. S. van der Zant, I. J. Ferrer, R. D'Agosta, A. Castellanos-Gomez, *Sci. Rep.* **2016**, *6*, 22214.
[42] K. Osada, S. Bae, M. Tanaka, H. Raebiger, K. Shudo, T. Suzuki, *J. Phys. Chem. C* **2016**, *120*, 4653.
[43] J. Dai, X. C. Zeng, *Angew. Chem., Int. Ed.* **2015**, *54*, 7572.
[44] J. Guo, J. Tao, Z. Zhang, L. Fei, D. Li, J. Jadwiszczak, X. Wang, Y. Guo, X. Liao, Y. Zhou, *ACS Appl. Electron. Mater.* **2020**, *2*, 3756.
[45] D. S. Muratov, A. R. Ishteev, D. A. Lypenko, V. O. Vanyushin, P. Gostishev, S. Perova, D. S. Saranin, D. Rossi, M. Auf der Maur, G. Volonakis, Feliciano Giustino, Per. O. Å. Persson, D. V. Kuznetsov, A. Sinitskii, A. Di Carlo, *ACS Appl. Mater. Interfaces* **2019**, *11*, 48021.
[46] A. Redon, *Electrochim. Acta* **1985**, *30*, 1365.
[47] Z. Tian, C. Han, Y. Zhao, W. Dai, X. Lian, Y. Wang, Y. Zheng, Y. Shi, X. Pan, Z. Huang, et al., *Nat. Commun.* **2021**, *12*, 2039.
[48] C. Wang, C. Zheng, G. Gao, *J. Phys. Chem. C* **2020**, *124*, 6536.
[49] A. Rathika, J. Peter, V. Ragavendran, J. Mayandi, P. Nithiananthi, et al., *Electrochim. Acta* **2024**, *477*, 143746.
[50] A. Pant, E. Torun, B. Chen, S. Bhat, X. Fan, K. Wu, D. P. Wright, F. M. Peeters, E. Soignard, H. Sahin, S. Tongay, *Nanoscale* **2016**, *8*, 16259.
[51] V. Q. Dang, K. Al-Ali, *New J. Chem.* **2020**, *44*, 7583.
[52] L. Huang, K. Tang, Q. Yang, G. Shen, S. Jia, *Mater. Res. Bull.* **2004**, *39*, 1083.
[53] X. Wang, K. Wu, M. Blei, Y. Wang, L. Pan, K. Zhao, C. Shan, M. Lei, Y. Cui, B. Chen, D. Wright, W. Hu, S. Tongay, Z. Wei, *Adv. Electron. Mater.* **2019**, *5*, 1900419.
[54] G. Wan, G. Zhang, X.-M. Lin, *Adv. Mater.* **2020**, *32*, 1905548.
[55] X. Li, X. Yang, Y. Huang, T. Zhang, B. Liu, *Adv. Mater.* **2019**, *31*, 1902031.
[56] S. Tomar, P. Sen, S. Chakraborty, *Sustainable Energy Fuels* **2022**, *6*, 5337.
[57] S. Tomar, S. Chakraborty, *Sustainable Energy Fuels* **2023**.
[58] T. Das, S. Chakraborty, R. Ahuja, G. P. Das, *ChemPhysChem* **2019**, *20*, 608.
[59] S. Santisouk, P. Sengdala, X. Jiang, X.-X. Xue, K.-Q. Chen, Y. Feng, *ACS omega* **2021**, *6*, 13124.
[60] M. Megha, P. Sen, *Phys. Chem. Chem. Phys.* **2023**.
[61] W. Wu, Y. Zhao, S. Li, B. He, H. Liu, X. Zeng, J. Zhang, G. Wang, *J. Colloid Interface Sci.* **2019**, *547*, 291.
[62] Y. Hu, H. Yu, L. Qi, J. Dong, P. Yan, T. Taylor Isimjan, X. Yang, *ChemSusChem* **2021**, *14*, 1565.
[63] A. Singh, S. Pakhira, *Energy Fuels* **2022**, *37*, 567.
[64] W. Kohn, L. J. Sham, *Phys. Rev.* **1965**, *140*, A1133.
[65] P. Hohenberg, W. Kohn, *Phys. Rev.* **1964**, *136*, B864.
[66] G. Kresse, J. Furthmüller, *Phys. Rev. B* **1996**, *54*, 11169.
[67] G. Kresse, J. Furthmüller, *Comput. Mater. Sci.* **1996**, *6*, 15.
[68] J. P. Perdew, K. Burke, M. Ernzerhof, *Phys. Rev. Lett.* **1996**, *77*, 3865.
[69] S. Grimme, *J. Comput. Chem.* **2006**, *27*, 1787.
[70] S. Grimme, J. Antony, S. Ehrlich, H. Krieg, *J. Chem. Phys.* **2010**, *132*, 154104.
[71] H. J. Monkhorst, J. D. Pack, *Phys. Rev. B* **1976**, *13*, 5188.
[72] Á. Morales-García, R. Valero, F. Illas, *J. Phys. Chem. C* **2017**, *121*, 18862.
[73] J. Heyd, G. E. Scuseria, M. Ernzerhof, *J. Chem. Phys.* **2003**, *118*, 8207.
[74] Y. Hinuma, G. Pizzi, Y. Kumagai, F. Oba, I. Tanaka, *Comput. Mater. Sci.* **2017**, *128*, 140.
[75] B. Mortazavi, F. Shojaei, M. Yagmurcukardes, M. Makaremi, X. Zhuang, *Energies* **2022**, *15*, 15.
[76] Y. Wu, M. Chan, G. Ceder, *Phys. Rev. B* **2011**, *83*, 235301.
[77] V. K. Yadav, S. H. Mir, J. K. Singh, *ChemPhysChem* **2019**, *20*, 687.
[78] J. Kang, S. Tongay, J. Zhou, J. Li, J. Wu, *Appl. Phys. Lett.* **2013**, *102*, 012111.
[79] A. Huang, W. Shi, Z. Wang, *J. Phys. Chem. C* **2019**, *123*, 11388.
[80] V. Chakrapani, J. C. Angus, A. B. Anderson, S. D. Wolter, B. R. Stoner, G. U. Sumanasekera, *Science* **2007**, *318*, 1424.
[81] J. K. Nørskov, T. Bligaard, A. Logadottir, J. R. Kitchin, J. G. Chen, S. Pandelov, U. Stimming, *J. Electrochem. Soc.* **2005**, *152*, J23.
[82] J. K. Nørskov, J. Rossmeisl, A. Logadottir, L. Lindqvist, J. R. Kitchin, T. Bligaard, H. Jónsson, *J. Phys. Chem. B* **2004**, *108*, 17886.
[83] R. Frydendal, M. Busch, N. B. Halck, E. A. Paoli, P. Krtil, I. Chorkendorff, J. Rossmeisl, *ChemCatChem* **2015**, *7*, 149.
[84] T. L. Tan, L.-L. Wang, D. D. Johnson, K. Bai, *Nano Lett.* **2012**, *12*, 4875.
[85] P. Liao, J. A. Keith, E. A. Carter, *J. Am. Chem. Soc.* **2012**, *134*, 13296.
[86] J. Rossmeisl, Z.-W. Qu, H. Zhu, G.-J. Kroes, J. K. Nørskov, *J. Electroanal. Chem.* **2007**, *607*, 83.
[87] C. Vayenas, S. Bebelis, S. Ladas, *Nature* **1990**, *343*, 625.
[88] J. Liu, *J. Phys. Chem. C* **2015**, *119*, 28417.

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